The Journal of Open Source Software

HighFidelityEphemerisModel.jl: a Julia package for High-fidelity astrodynamics

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Software

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Summary

A description of the high-level functionality and purpose of the software for a diverse, non-specialist audience.

Statement of need

A section that clearly illustrates the research purpose of the software and places it in the context of related work. This should clearly state what problems the software is designed to solve, who the target audience is, and its relation to other work.

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State of the field

Notable open-source packages that include HFEM dynamics include the following:

Orekit (Maisonobe et al., 2010)

GMAT (Hughes et al., 2014)

Basilisk (Kenneally et al., 2020)

Nyx

Brahe

Software design

An explanation of the trade-offs you weighed, the design/architecture you chose, and why it matters for your research application. This should demonstrate meaningful design thinking beyond a superficial code structure description.

Research impact statement

Evidence of realized impact (publications, external use, integrations) or credible near-term significance (benchmarks, reproducible materials, community-readiness signals). The evidence should be compelling and specific, not aspirational.

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AI usage disclosure

Transparent disclosure of any use of generative AI in the software creation, documentation, or paper authoring. If no AI tools were used, state this explicitly. If AI tools were used, describe how they were used and how the quality and correctness of AI-generated content was verified.

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References

Hughes, S. P., Qureshi, R. H., Cooley, D. S., & Parker, J. S. (2014). Verification and validation of the General Mission Analysis Tool (GMAT). *AIAA/AAS Astrodynamics Specialist Conference*, 4151.

Kenneally, P. W., Piggott, S., & Schaub, H. (2020). Basilisk: A flexible, scalable and modular astrodynamics simulation framework. *Journal of Aerospace Computing, Information, and Communication*, *17*(9), 496–507. https://doi.org/10.2514/1.I010762

Maisonobe, L., Parraud, P., & Vehil, R. (2010). Orekit: An open source library for operational flight dynamics applications. *Proceedings of the 4th International Conference on Astrodynamics Tools and Techniques (ICATT)*.

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